

Physical Sciences & Engineering Ethics Review and Approval Form for UG & PGT students



IMPORTANT NOTE: Research projects cannot begin until ethical approval has been granted.

Please complete the relevant sections of this form if, after filling out the relevant ethical review checklist (Annex A), you have identified a potential ethical issue. **Please submit a copy of the completed Annex A as part of your ethics application.**

If required, contact your Project Supervisor to obtain further guidance prior to the completion of this form.

Please refer to the ethical review procedures outlined in the Project Guidelines or contact the Course or Programme Coordinator for further advice on the application process.

Information	Response
Name:	Fraser McCarter
ID number:	52208648
School:	School of Natural and Computing Sciences
Department or discipline:	Computing Science
Programme (e.g., BEng, MSc):	MA
Project Title: Course Number and Name:	Prototyping a Python-Based Engine for Music Scripting and Algorithmic Composition
Names of other individuals involved in the research/project?	
Name and email address of main supervisor:	Dr Debbie Meharg debbie.meharg@abdn.ac.uk
Signature of main supervisor:	
Application date:	17/03/2026

Please note:

1. Research involving NHS staff, patients, facilities and premises may be subject to ethical review by an NHS ethics committee. This includes research involving individuals when their status as NHS staff or patients is relevant to the research, even when a medical condition is not the subject of the research. Research involving adults who do not have the capacity to consent may also be subject to these ethical review procedures. Further guidance is available at researchgovernance@abdn.ac.uk
2. Research involving animal and biological materials may be subject to additional approval requirements. Please contact your Supervisor for further guidance.

3. Research involving the collection of genetic resources (organisms, microorganisms, DNA, RNA, proteins, small molecules) from signatories to the Convention on Biodiversity/Nagoya Protocol requires a formal agreement to be in place before this research can begin. Contact your [Research Development Executive](#) for further guidance.

CHECKLIST

The purpose of this checklist is to make sure no information has been inadvertently left out and to allow reviewers to assess the application more quickly. **If you do not complete the checklist and attach a completed Annex A (and any other documentation where required), the application will be returned to you, hence delaying the approval process.**

I confirm that if my project changes significantly then I will notify the Course/Programme Coordinator. **YES/NO**

I have **attached** a completed checklist (Annex A). **YES/NO**

I confirm that I have discussed this application with my supervisor. **YES/NO**

This project requires me to **travel outwith the UK** **YES/NO**

If YES, please provide the following confirmation:-

- I will comply with the requirements of the University's [Overseas Travel Policy](#), including obtaining permission to travel (where required by the policy), completion of a [risk assessment](#) and will obtain [University travel insurance cover](#).

YES/NO

Other Attachments (delete YES/NO/NOT APPLICABLE as appropriate):

I have attached a Participant Information Sheet. YES

I have attached a Consent Form YES

I have attached a schedule of questions for surveys and/or interviews. YES

<u>Project description</u> <i>Please attach a project descriptor or summary document (where available)</i>	A participant-based usability survey conducted using microsoft forms. The survey looks to gauge user interaction with the prototype music scripting engine through three tasks. Users will give feedback on these tasks though likert scale and open ended questions. Two versions of the survey are available to complete, an interactive one, in which participants download a copy of the code from GitHub: https://github.com/frasermccarter/CS4525-Joint-Honours-Project (code is fully inspectable to the participant before download), or through watching a video demonstration of the completion of these tasks.
<u>Project start date and duration</u> (i.e. the particular piece of work for which you are applying for ethical approval [not your overall programme of research])	Start date: 23/03/2026 Duration: 23/03/2026 - 01/05/2026
<u>Project methodology</u>	The survey will be handed out to eligible potential participants, this will be done digitally through a link to the microsoft forms page for the survey. Further instructions and information about the survey will be given on the first page, before a potential participant decides whether to participate or not. The following page will ask for potential participants consent and willingness to continue with the survey. If so, the participant continues on to the survey questions. The survey is split into five sections: Programming and Musical Background - asks users about their current background and familiarity with both fields. After completing this the participant is then given a choice of whether they wish to complete the three upcoming tasks interactively or through watching and commenting on a video demonstration. Task one, two and three ask participants to engage with the program and evaluate the main core features, giving insight and feedback through likert scale questions and the opportunity to expand on any points through a text box if they so wish. Task One - If the interactive version of the survey is chosen: ‘In a separate tab please open the link below, this will take you to the GitHub repository for the code. Download the code, extract it and open the folder in your chosen coding environment. Each task will be labelled with its own file in the side tab (task1.py, task2.py, etc.). Please select task1.py for this task. https://github.com/frasermccarter/CS4525-Joint-Honours-Project Task One - Find the Eb note and change it to an E, then run the code to play the result. - Now change the octave of the G note from an G3 to a G4, then run the code to play the result. - Change the duration of the C note from 1 to 0.5, then run the code to play the result. - Create a new A4 note (with a duration and velocity of your choosing) that plays after the G note, then run the code to play the result.’

	OR if the video demonstration version of the survey is chosen: ‘Please watch the video demo for task one below. Try to imagine as though you yourself are using the program and think about what it would be like to complete this task and interact with the system. The task at hand is shown in the video as well as listed below: https://youtu.be/8FCgav_1HQU Task One - Find the Eb note and change it to an E, then run the code to play the result. - Now change the octave of the G note from an G3 to a G4, then run the code to play the result. - Change the duration of the C note from 1 to 0.5, then run the code to play the result. - Create a new A4 note (with a duration and velocity of your choosing) that plays after the G note, then run the code to play the result.’ Task Two - If the interactive version of the survey is chosen: ‘In a separate tab please open the link below, this will take you to the GitHub repository for the code. Download the code, extract it and open the folder in your chosen coding environment. Each task will be labelled with its own file in the side tab (task1.py, task2.py, etc.). If you have already done this step then ignore and continue. Please select task2.py for this task. https://github.com/frasermccarter/CS4525-Joint-Honours-Project Task Two - Locate the arpeggio function and change the chord from a C major (Cmaj) to a C minor (Cmin), then run the code to play the result. - Change the arpeggio from playing an ascending sequence (up) to a descending sequence (down), then run the code to play the result. - Fill in the parameters for the random walk function, then run the code to play the result. You must fill in: - start_pitch: The starting pitch of the random walk (e.g. 'C4', 60 (midi), 440.0 (hz) etc.). - num_notes: The total number of notes to generate in the random walk (e.g. 4, 5, 16 etc.). - max_step: The maximum step size (in semitones) for the random walk. A max_step of 2 means that each note can be at most 2 semitones apart from each other. - duration: The duration of each note in beats (e.g. 0.5 for quaver, 1 for crochet, etc.). - scale_type: The scale to constrain the random walk to (e.g. "major", "minor", "chromatic"). If left unspecified (scale_type=None), the random walk will be generated without any scale constraints, meaning it can include any pitch.’ OR if the video demonstration version of the survey is chosen: ‘Please watch the video demo for task two below. Try to image as though you yourself are using the program and think about what it would be like to complete this task and interact with the system. The task at hand is shown in the video as well as listed below: https://youtu.be/2kGDQyuzPtc
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Task Two

- Locate the arpeggio function and change the chord from a C major (Cmaj) to a C minor (Cmin), then run the code to play the result.
- Change the arpeggio from playing an ascending sequence (up) to a descending sequence (down), then run the code to play the result.
- Fill in the parameters for the random walk function, then run the code to play the result.

You must fill in:

- start_pitch: The starting pitch of the random walk (e.g. 'C4', 60 (midi), 440.0 (hz) etc.).
- num_notes: The total number of notes to generate in the random walk (e.g. 4, 5, 16 etc.).
- max_step: The maximum step size (in semitones) for the random walk. A max_step of 2 means that each note can be at most 2 semitones apart from each other.
- duration: The duration of each note in beats (e.g. 0.5 for quaver, 1 for crochet, etc.).
- scale_type: The scale to constrain the random walk to (e.g. "major", "minor", "chromatic"). If left unspecified (scale_type=None), the random walk will be generated without any scale constraints, meaning it can include any pitch.'

Task Three - If the interactive version of the survey is chosen: 'Participants are not required to create an account or provide personal data on third-party platforms. Though not expected, this task may be skipped if the participant feels frustrated at any point.

In a separate tab please open the link below, this will take you to the GitHub repository for the code. Download the code, extract it and open the folder in your chosen coding environment. Each task will be labelled with its own file in the side tab ([task1.py](#), [task2.py](#), etc.). Please select [task3.py](#) for this task.

<https://github.com/frasermccarter/CS4525-Joint-Honours-Project>

Task Three

- Run the code and listen to its musical output.
- Export the musical output of the Python script as a MIDI file using:
export_midi("task3_output", tempo=120)
- Once exported, upload the MIDI file to the online MIDI editor / player linked below:

<https://signalmidi.app/edit>

Please listen and compare the results to what you expect based on the code input and whether the MIDI output is an accurate representation of the code.'

OR if the video demonstration version of the survey is chosen: 'Please watch the video demo for task three below. Try to imagine as though you yourself are using the program and think about what it would be like to complete this task and interact with the system. The task at hand is shown in the video as well as listed below:

	https://youtu.be/Yf90T9oSniI Task Three - Run the code and listen to its musical output. - Export the musical output of the Python script as a MIDI file using: export_midi("task3_output", tempo=120) - Once exported, upload the MIDI file to the online MIDI editor / player linked below: https://signalmidi.app/edit Please listen and compare the results to what you expect based on the code input and whether the MIDI output is an accurate representation of the code. Overall System Review - In this section participants give feedback on the program as a whole rather than on a task-based level, questions consist of likert scale and open ended prompts. Once complete participant answers are stored on microsoft forms and all answers are anonymised from point of submission.
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Section 1: Research projects involving human participants (not NHS staff or patients)

If you answered 'No' to Q1 and Q2 then omit Q3 – 18 and proceed straight to Q19.

Recruitment Procedures

	Question	Yes	No
1	Does your project involve human participants? This includes use of surveys, questionnaires, on-line surveys and tests, focus groups and workshops where human participants provide information or data to inform the research.	X	
2	(a) Does your project involve human remains? (b) If so, does your work conform with the Historic Environment Scotland guidelines?		X
3	Does your project involve people less than 18 years of age ¹ ?		X
4	Does your project involve people with learning or communication difficulties?		X
5	Is your project likely to involve people involved in illegal activities?		X
6	Does your project involve people belonging to a vulnerable group, other than those listed above?		X

¹ It is University policy that children under the age of 16 should also have parental consent to participate in research projects, in addition to consent obtained from the child. Research participants aged 16 and over in Scotland are not required to have parental consent to participate in research projects. However in most countries it is 18 years and in these circumstances, parental consent will also be required for participants aged 17 or under. If you are working with research participants between the ages of 16 – 18 it is essential that you check the local requirements regarding parental consent.

	Question	Yes	No
7	Does your project involve people with whom you have, or are likely to have, a working or professional relationship: for instance, staff or students of the university, professional colleagues or clients?	X	
8	Does your project involve people who do not have English as their first language?		X
9	Does your project require the recording of audio or video of participants or of others not involved in the research?		X
10	Do you plan to conceal your own identity during the course of your project?		X

The online survey and accompanying online tasks will be made available to people with whom I have, or are likely to have, a working or professional relationship as they are likely to meet the inclusivity criteria and no predetermined biases would exist (based on relationship) that could affect survey results. The survey will recruit between 25-30 participants. The survey will be handed out to eligible potential participants, this will be done digitally through a link to the microsoft forms page for the survey.

If you answered 'Yes' to any of the above questions, please provide further details.

If you answered 'Yes' to Q1 then you must provide a Participant Information Sheet and a Consent Form. For web-based research, screenshots of the appropriate web pages suffice.

If your project involves surveys or interviews then you must provide a schedule of questions.

If you answered 'Yes' to Q3, Q4 or Q6 then you will need to apply for disclosure through Disclosure Scotland if you intend to be alone with a research participant or have to take sole responsibility for the participants at any point during your research activity.

**Consent
Procedure
s**

	Question	Yes	No
11	Do you have set procedures that you intend to use for obtaining informed consent from all participants, including (where appropriate) parental consent for children?	X	
12	Will you tell participants that their participation is voluntary?	X	
13	Will you obtain written consent for participation, including for audio and/or video recording?	X	
14	Will you tell participants that they may withdraw from the research at any time and for any reason?	X	
15	Will you give potential participants a period of time to consider participation?	X	
16	Does your project involve concealment or deliberately misleading participants?		X

If you answered 'Yes' to any of the above questions, please provide further details.

Procedures that will be taken are outlined in the User Participation and Online Consent documents. Users will have to accept the terms and conditions of the Survey before participating. This involves reading through the information provided in the Online Consent document.

Possible Harm to Participants

	Question	Yes	No
17	Does the research have the potential to cause distress or discomfort to participants or any member of the research team?		X
18	Is there any realistic risk of any participants experiencing a detriment to their interests as a result of participation?		X

If you answered 'Yes' to either of the above questions, please provide further details. Explain how this risk was assessed and how you propose to manage it.

Section 2: Data protection, handling and storage

IMPORTANT NOTE:

The General Data Protection Regulation imposes a number of obligations for the use of **personal data** (defined as any information relating to an identified or identifiable living person) or including the use of personal data in research.

If you are using personal data, you should consider whether your research requires a [Data Protection Impact Assessment](#) (DPIA) and complies with the University Data Protection policy.

If you are, you now need to see the [Data Protection Checklist for Researchers](#) for guidance.

If you then feel that a DPIA may be required or you need data protection advice, then you should contact the Data Protection Officer dpa@abdn.ac.uk.

Please provide the following confirmation:

I have read the above guidance and have met the relevant data protection obligations.

☒ **Please tick the box to confirm**

In addition, you should also check the requirements for a Data Management Plan (DMP) in the [Research Data Management Policy](#) and [Guidance](#).

Once checked, please confirm the requirement by ticking one of the following:

☒ **No requirement for DMP**

☐ **DMP required and this is attached**

Please see [here](#) for guidance on creating a DMP. For further support, contact digitalresearch@abdn.ac.uk

	Question	Yes	No
19	(a) Will any non-anonymised and/or personal data be generated and/or used? (b) Will you use an existing dataset in your research? (c) If 'yes', do you have permission to do so?		X
20	Will any data be stored (temporarily or permanently) anywhere other than on password-protected University computers or servers?	X	
21	Will you gain access to sensitive ² data about living individuals or organisations that is not already publicly available elsewhere ? If 'yes', will you gain the consent of the individuals concerned?		X
22	Does your project require access to personal data about participants from other parties (e.g., teachers, employers), databanks or files? <i>If 'yes', please explain in the comments section below how you will gain the consent of these participants.</i>		X
23	Does the project involve collecting personal data from websites or from social media (e.g., Facebook, Twitter)?		X
24	Will the data be stored, collected or accessed from: - outside the UK? - outside the EU?		X
25	Is the data likely to contain material that is indecent, offensive, defamatory, threatening, discriminatory or extremist? <i>If yes, see here for an explanation of the obligations of the researcher and the university under the Prevent duty.</i>		X
26	Are there any contractual conditions attached to working with or storing the data? (E.g., an HSCIC data sharing agreement.)		X
27	Could working with this data damage the University's reputation? (E.g., bad press coverage, public protest.)		X
28	Could working with this data cause an increased risk of attack (cyber- or otherwise) against the University? (E.g., from pressure groups.)		X

For further advice on Data Protection and GDPR, please refer to www.abdn.ac.uk/dataprotection.

If you answered 'Yes', to any of the above questions, please provide further details. Explain how you intend to ensure that data is stored securely and in line with the requirements of the Data Protection Act and funding bodies (if

² Sensitive data includes data that relates to racial or ethnic origin, political opinions, religious beliefs, trade union membership, physical or mental health, sexual life, actual and alleged offences.

The survey will take place on Microsoft Forms, of which participants will be informed. I will personally make sure that Microsoft's data protection policy aligns with the Data Protection Act.

<https://support.microsoft.com/en-gb/office/security-and-privacy-in-microsoft-forms-7e57f9ba-4aeb-4b1b-9e21-b75318532cd9>

All raw data will be accessed only by the researcher and the project supervisor for the purposes of analysis and academic reporting. The data will be anonymised at the point of collection by assigning participants a unique identifier and removing any identifying information from responses. Once anonymised, it will not be possible to identify participants using any reasonably available means.

applicable). Give specific consideration to whether any non-anonymised and/or personal data will be generated and/or stored and what precautions you will put in place regarding access.

Section 3: Research involving possible harm to the environment

	Question	Yes	No
29	Does the project have the potential to cause environmental damage or harm? <i>This includes the natural environment but also buildings and structures created by people, especially ones of historical or archaeological importance.</i>		X

If you answered 'Yes' to the above question, please provide further details. Explain how this risk was assessed and how you propose to manage it. Say whether relevant guidelines exist in your discipline, and whether you intend to follow them.

Section 4: Research which may have an adverse impact on national security

	Question	Yes	No
30	Does your project give rise to a realistic risk to the national security of any country?		X

If you answered 'Yes' to the above question, please provide further details. Explain how this risk was assessed and how you propose to manage it.

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Section 5: Funding and conflict of interest

	Question	Yes	No
31	Is your project funded by the university or an outside organisation, or have you applied for funding?		X
32	Is there any potential conflict of interest between research funder and researchers or participants and researchers which may potentially affect the research outcome or the dissemination of research findings?		X
33	Might the project lead to financial gain to funders, investigators or participants?		X

If you answered 'Yes' to any of the above questions, please provide further details. Explain any potential conflict of interest and how you propose to manage it.

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For any contractual or intellectual property questions, please contact the Research Development team in Research & Innovation (june.middleton@abdn.ac.uk).

Incomplete or incorrectly completed forms will be returned to the applicant, delaying the process of obtaining ethics approval. Make sure you have completed the checklist on page 2.

Please refer to the ethical review procedures outlined in your Project Guidelines or contact your Course / Programme Coordinator for further advice.